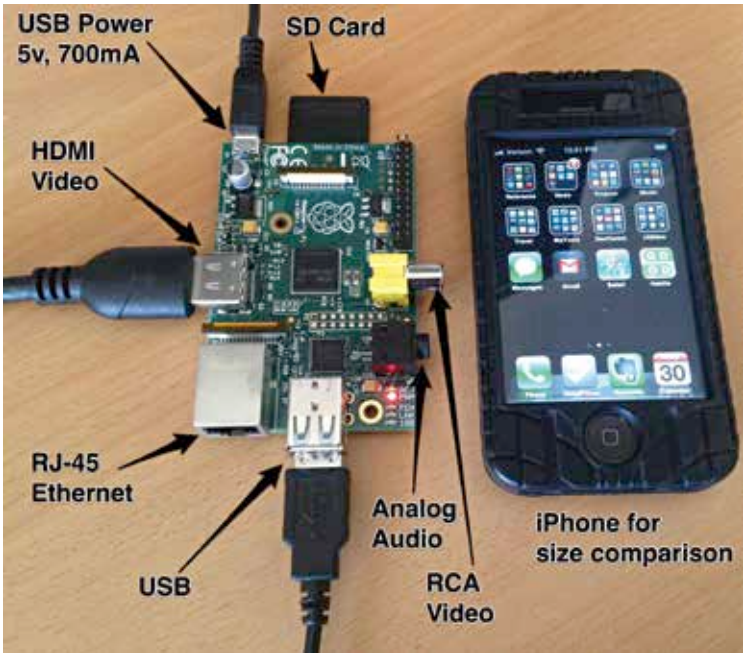




An example of I/O devices connected to the RPi

Application of SoCs

Since the 80s, if a device is small and battery powered, there are major chances that it is powered by an SoC. Smartphones, tablets, even the Nintendo Switch and the PS Vita; everything is run on these powerful mini-monsters. And they are indeed monsters when it comes to processing power. For example, the Microsoft Surface Tab was powered by Tegra 3 SoC. Again, the power and processing of an SoC depends entirely on how much processing it was built to handle i.e. the task that the chip is expected to execute.

Besides CPU, GPU and memory, an SoC can also host another important component called a Northbridge, which handles communications between CPU and other components of SoC. This gives the SoC the ability to include features like cellular transmission, 4G, Bluetooth etc. Most SoCs are powered by ARM processors. They are chosen as they have the suitable architecture to deliver high performance with low power consumption. The major drawback of SoCs is the cost involved. Although simple SoCs